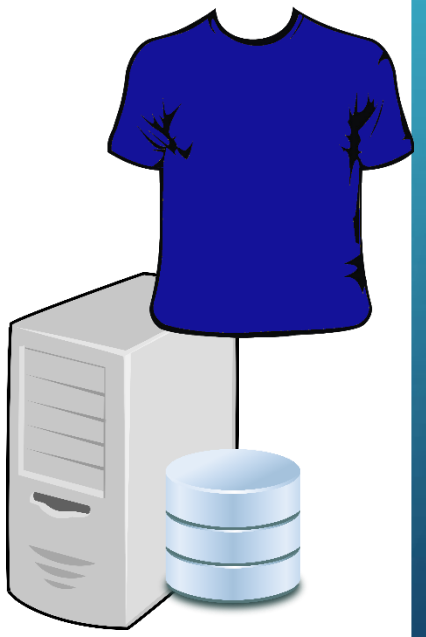
COMP 3920

DATABASE SYSTEMS 1

PATRICK GUICHON

T-SHIRT WEBSITE

Web Server



Database

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T-SHIRT WEBSITE - FAILURE

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T-SHIRT WEBSITE - SOLUTION

INSERT

UPDATE

Web Server

Database

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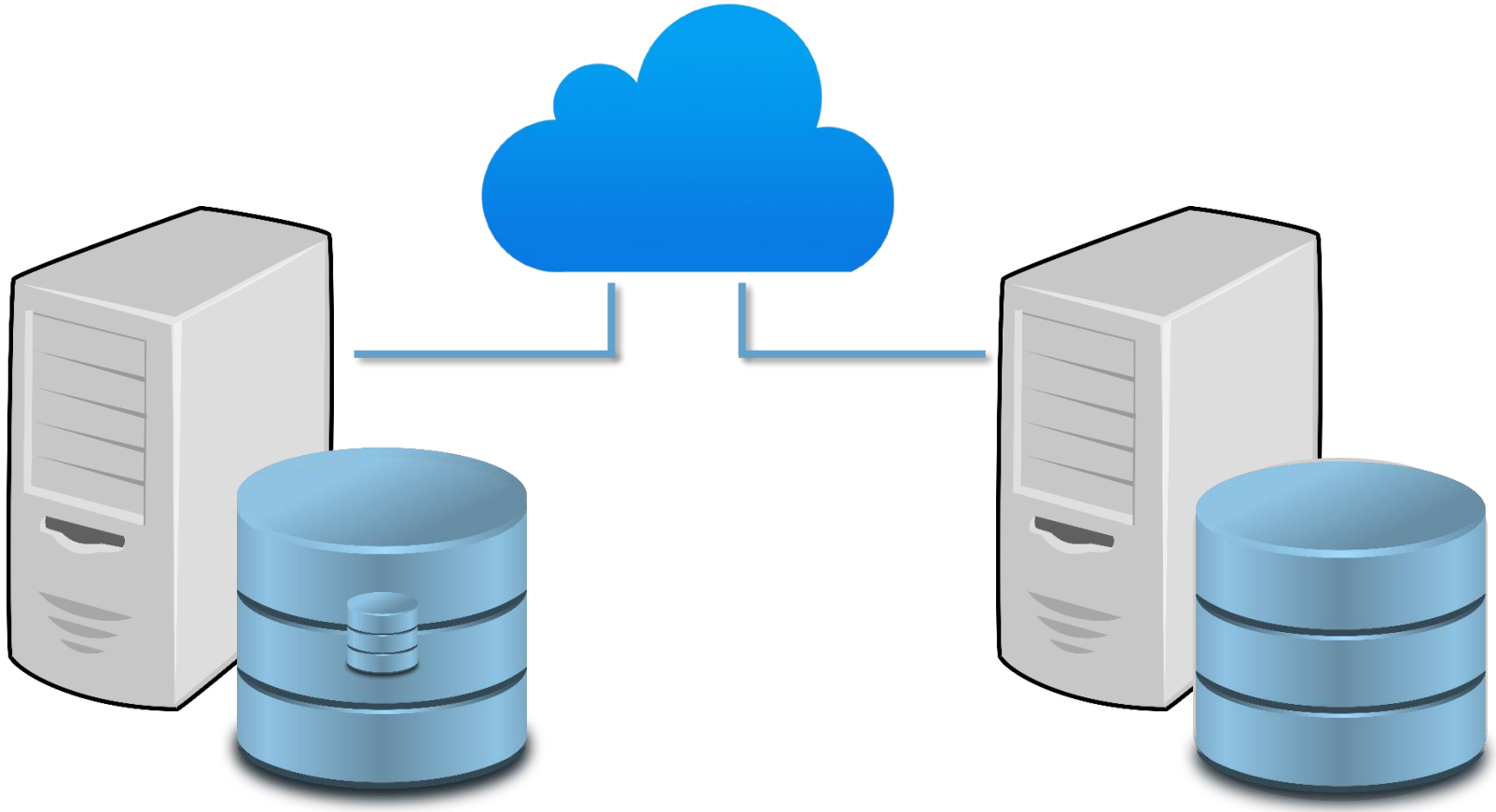
REPLICATION



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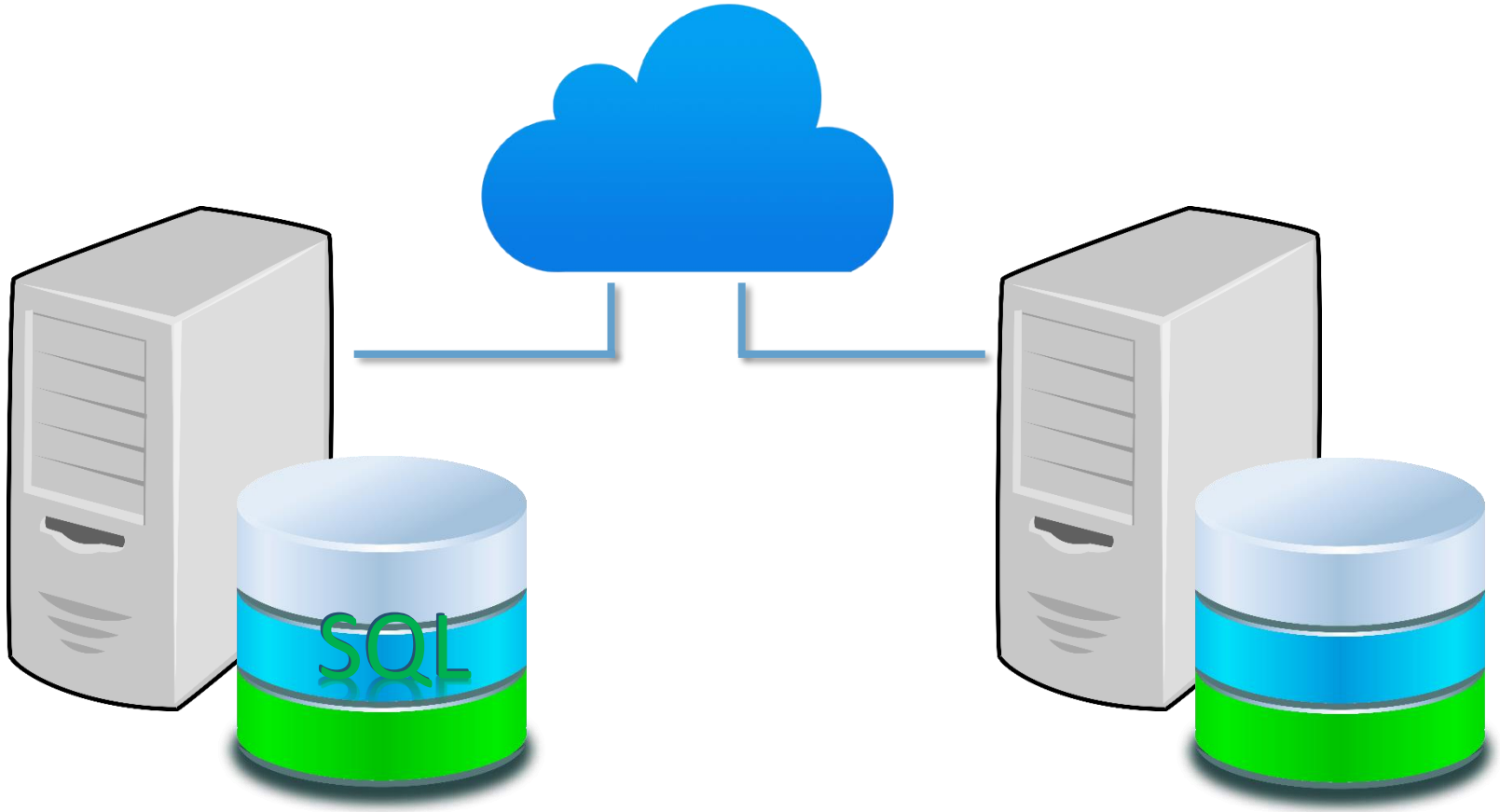
REPLICATION



Master

Slave

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REPLICATION

- Why use replication?
  - Great way to achieve high availability (uptime)
    - Disaster Recovery
    - Reduce Latency
  - Scalability
  - Data Warehousing
  - Test/Development Servers
  - Reporting

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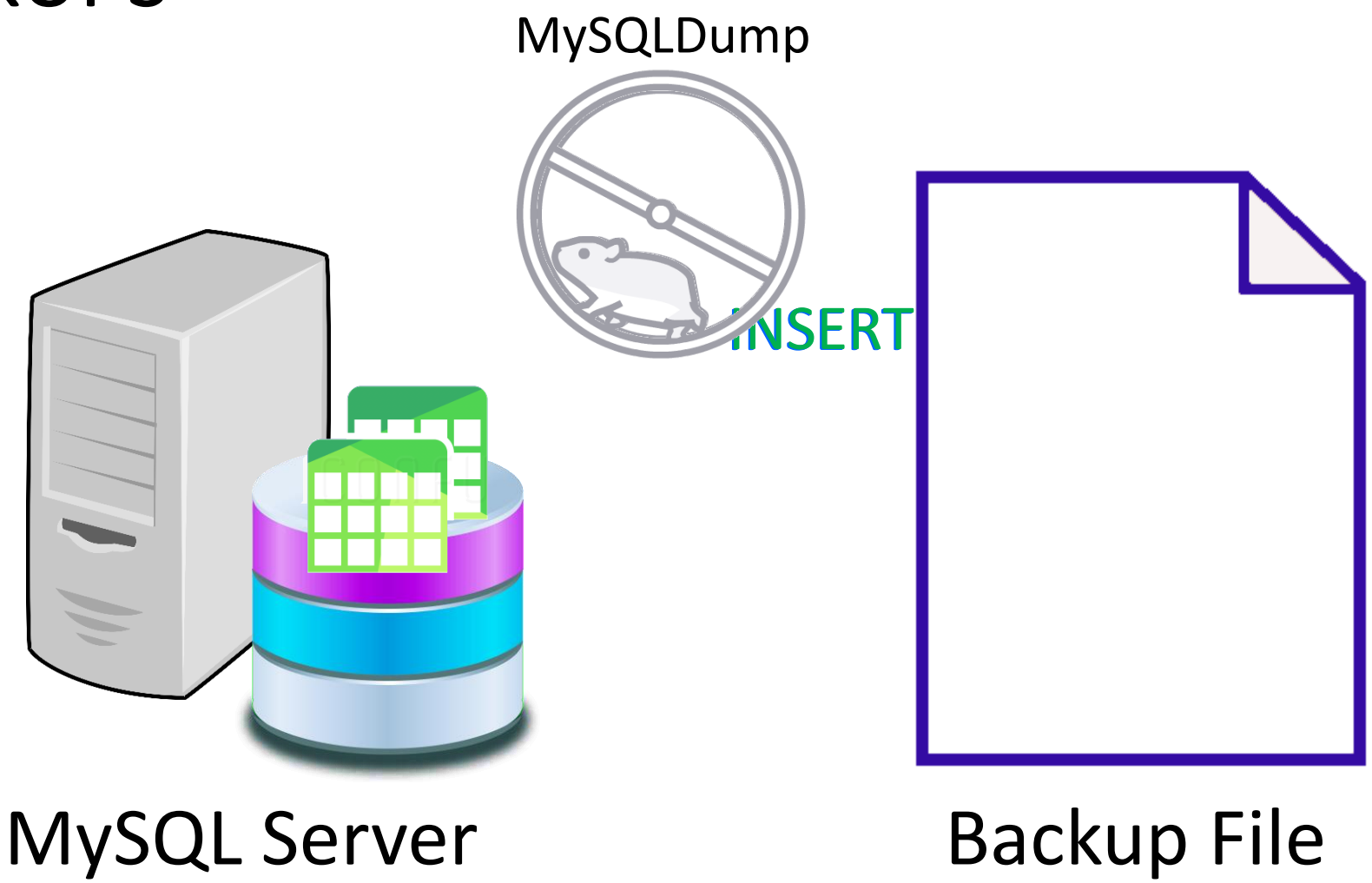
BACKUPS



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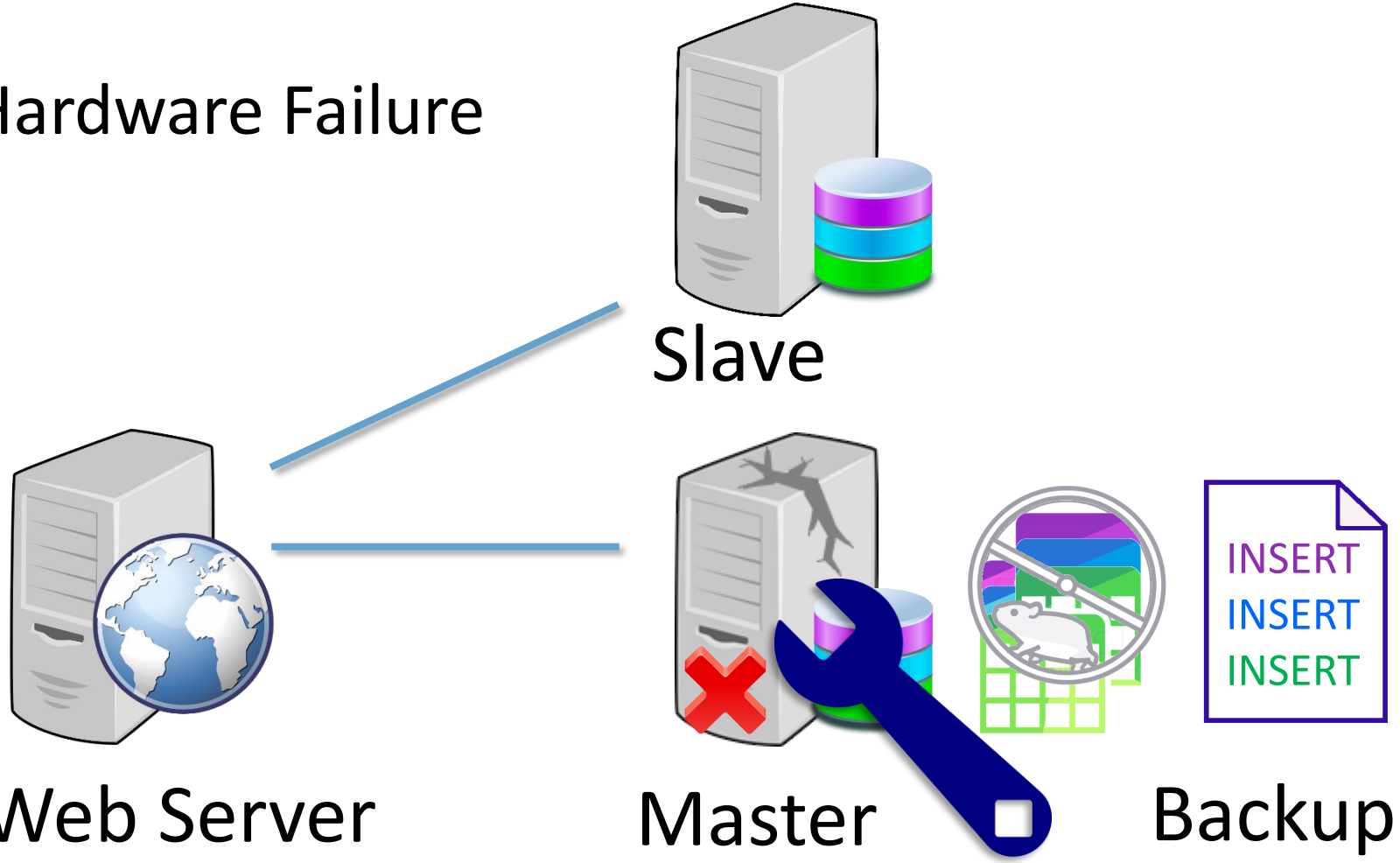
BACKUPS VS REPLICATION #1

Scenario 1 - Hardware Failure



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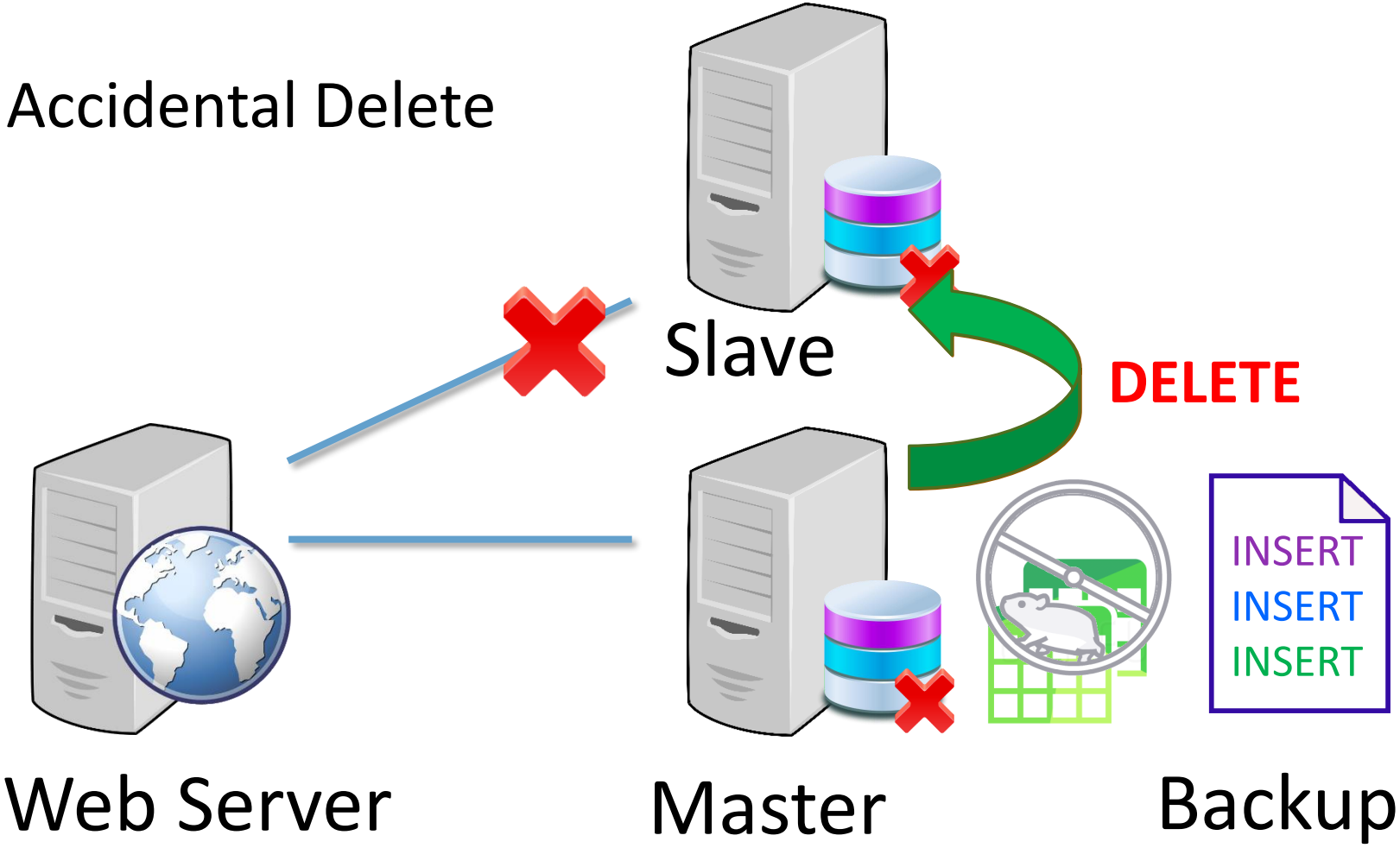
BACKUPS VS REPLICATION #2

- Scenario 2 - Accidental Delete

DELETE



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SUMMARY

	Recovery From	
	Failed Hardware	Accidental Delete
Replication		
Backup		
Replication with Backups	Fast ✔	Yes ✔

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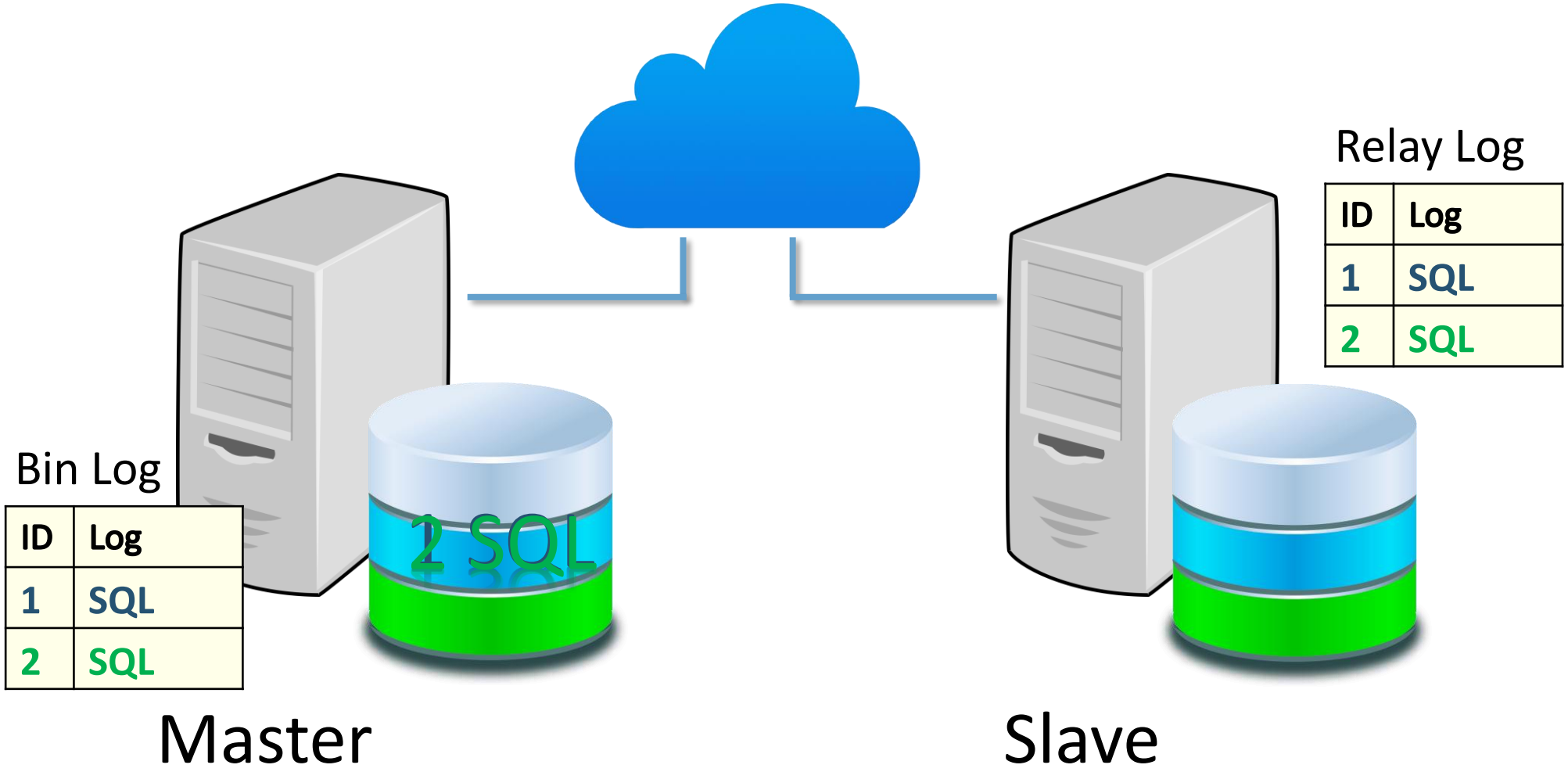
REPLICATION



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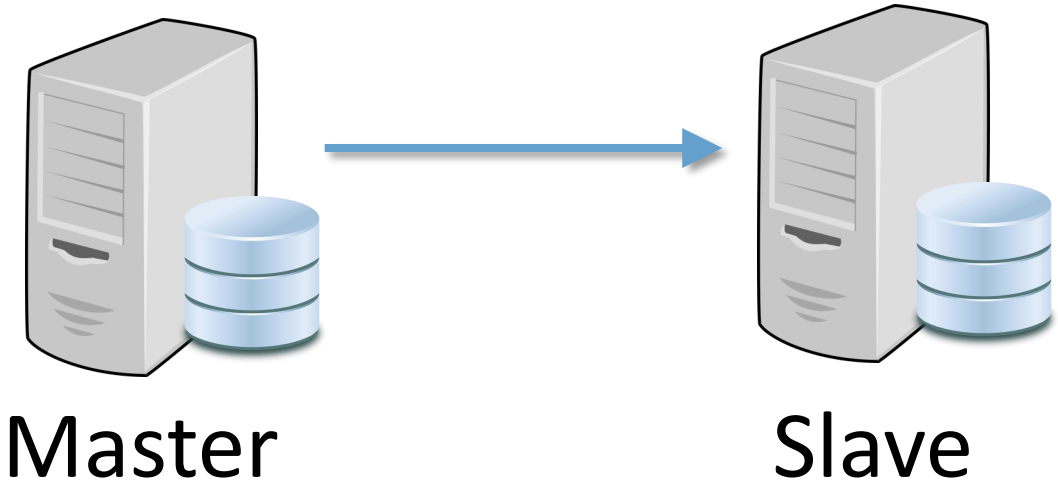
REPLICATION SCENARIOS

- 1 Master ⇨ 1 Slave



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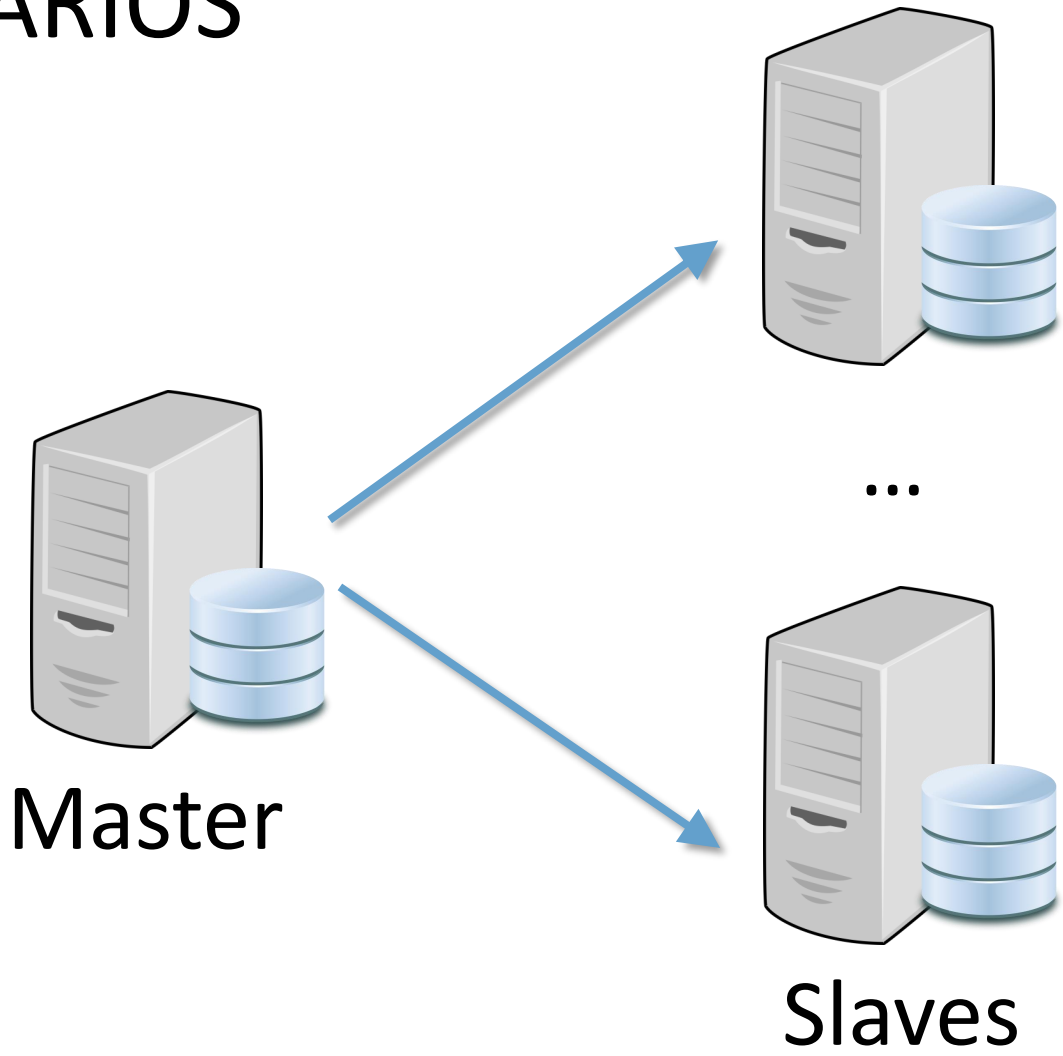
REPLICATION SCENARIOS

- 1 Master ⇨ * Slaves



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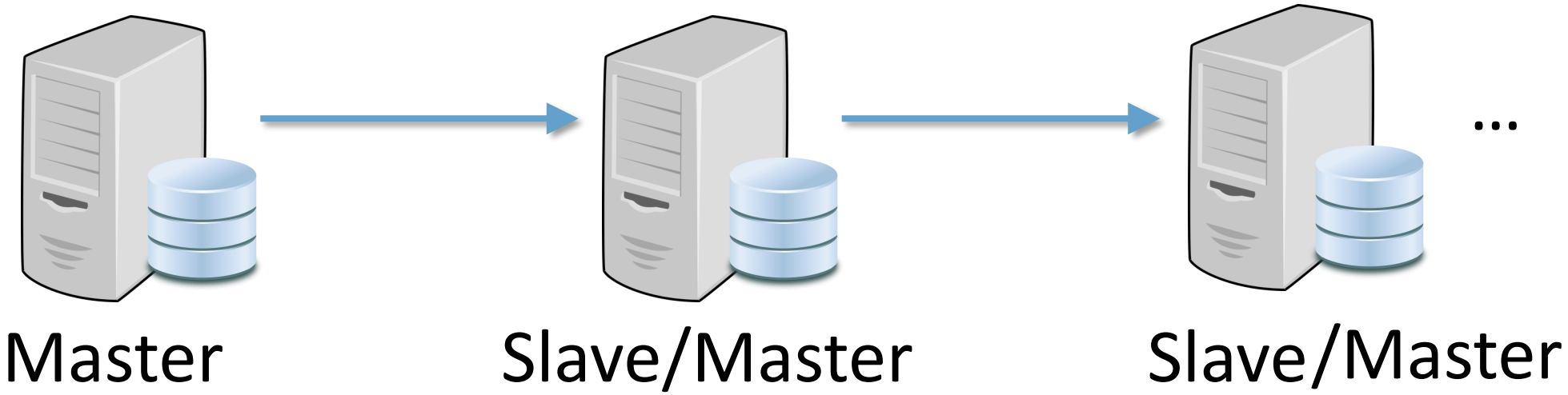
REPLICATION SCENARIOS

- 1 Master ⇨ 1 Slave/Master ⇨ 1 Slave/Master ⇨ …



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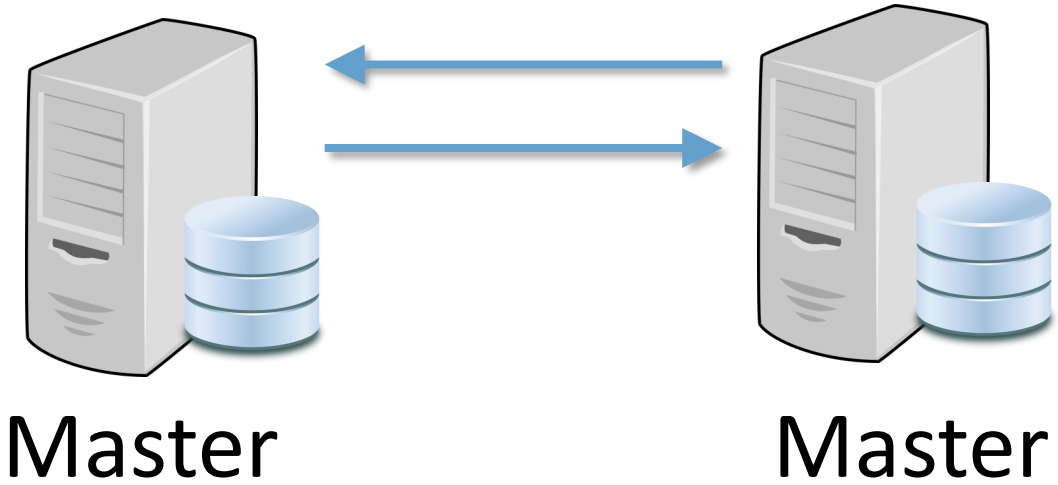
REPLICATION SCENARIOS

- 1 Master ⇔ 1 Master



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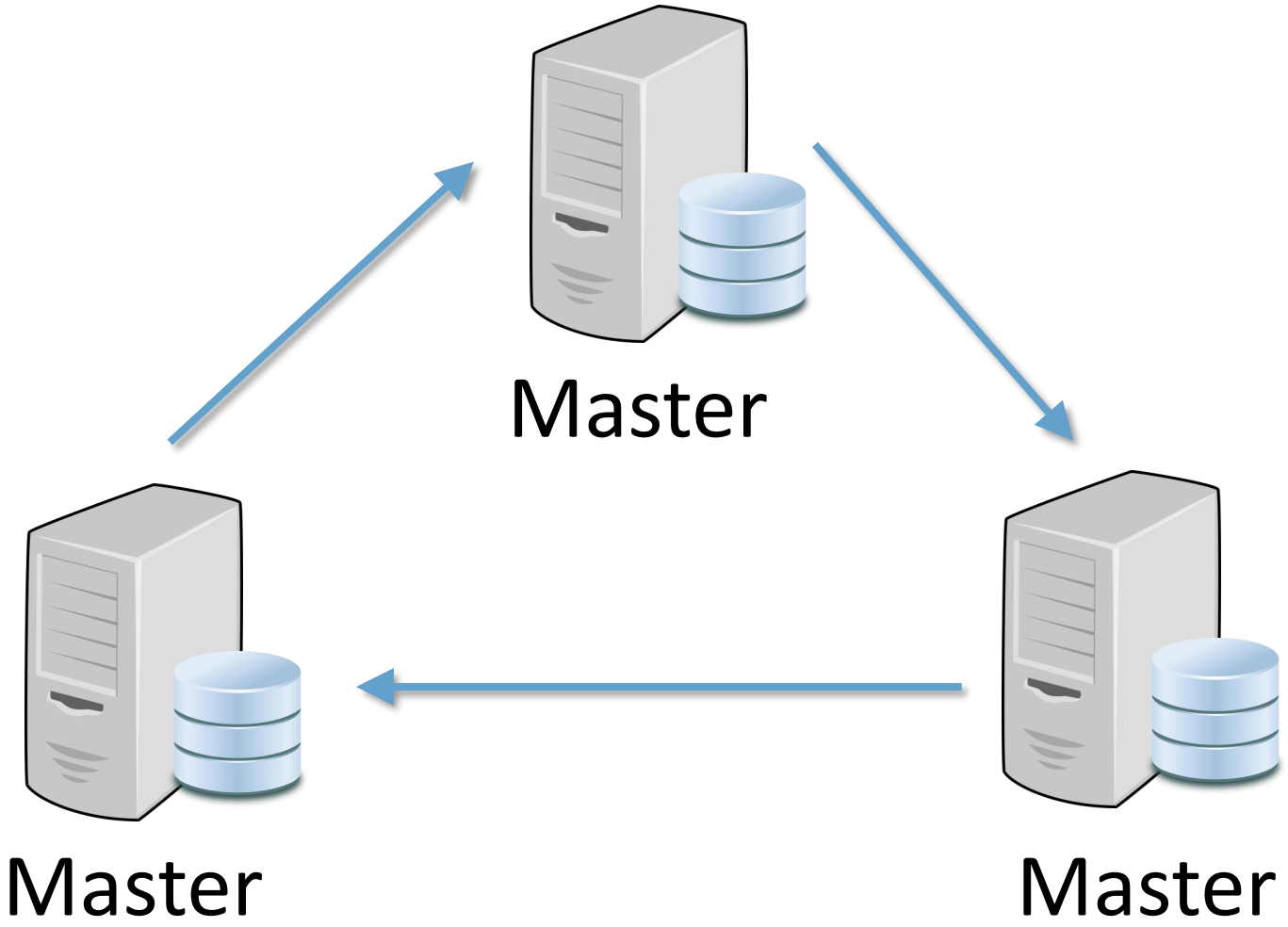
REPLICATION SCENARIOS

- *Master Ring



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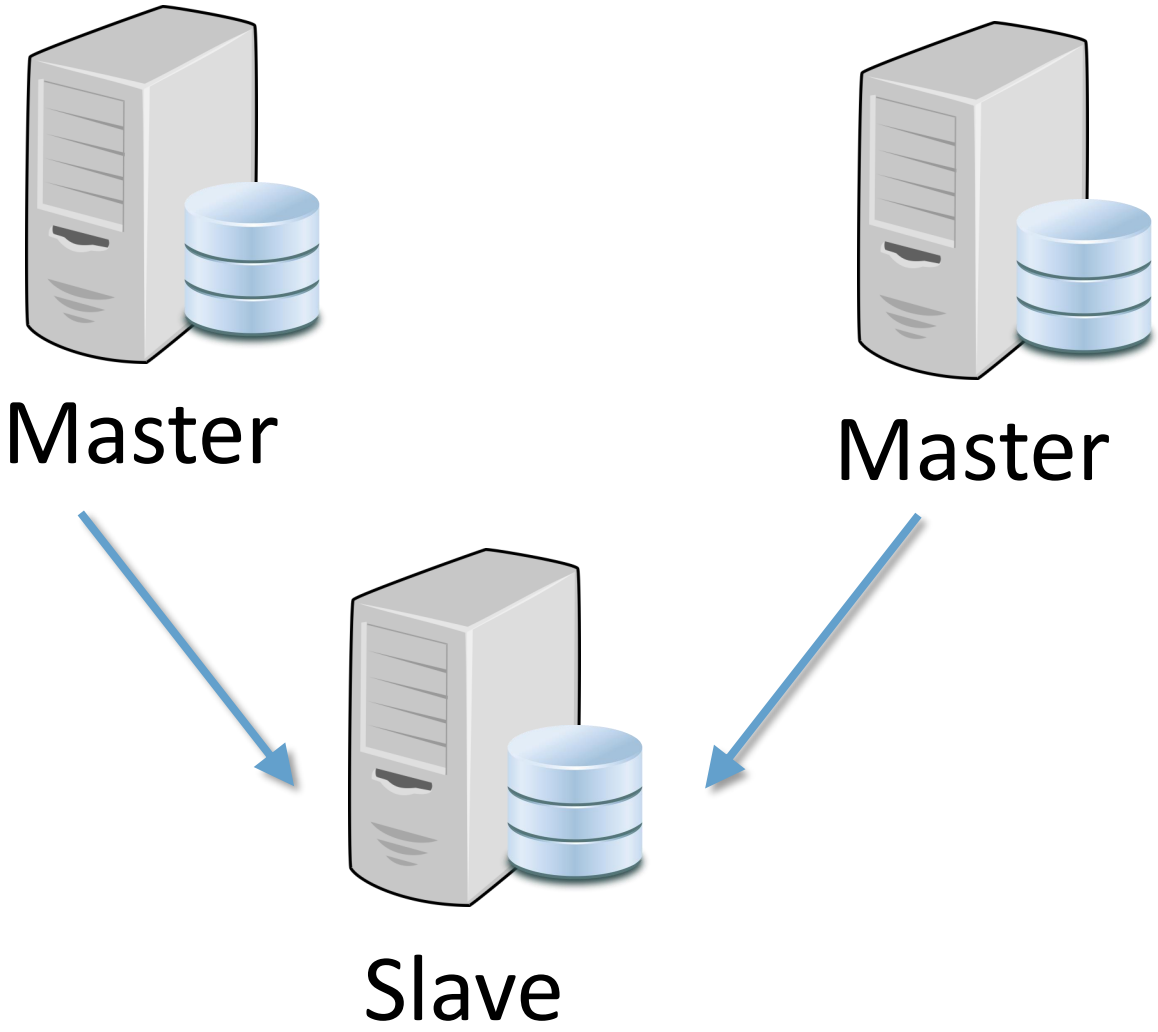
REPLICATION SCENARIOS

- Slave with
  2 Masters

- Allowed
  as of MySQL 5.6
  Use ‘`channels`’.



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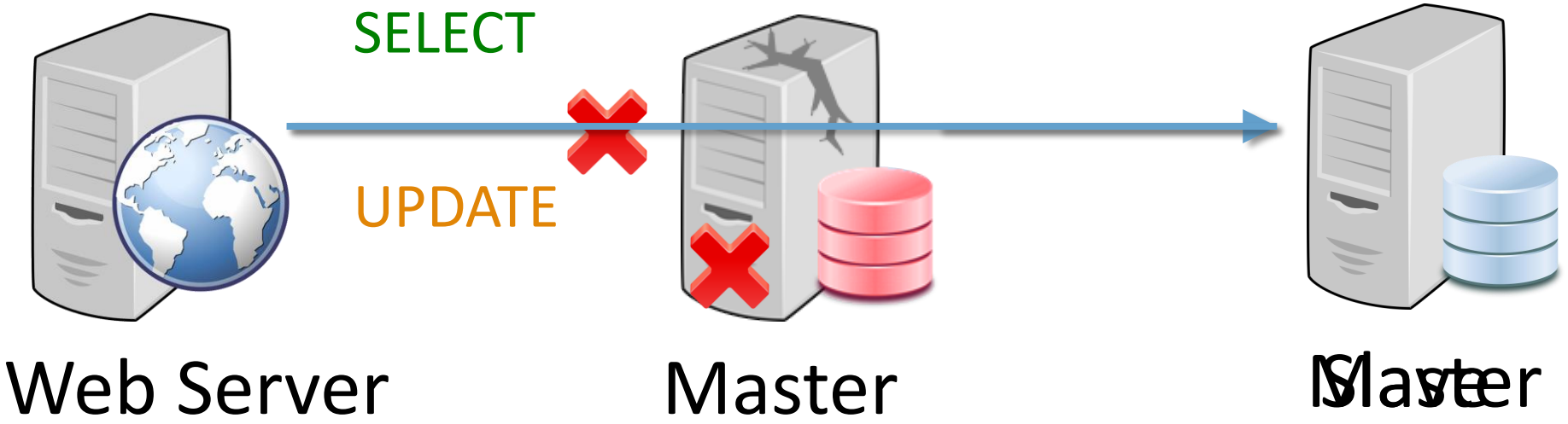
REPLICATION REQUIREMENTS

- Compatible Master/Slave Servers
- Binary Logs Enabled on Master
- Relay Logs Enabled on Slave
- Log Slave Updates (if you want the Slave to be a Master)
- Unique Master/Slave Server IDs
- Replication User – used to read Master’s Binary Log

COMPATIBLE MASTER/SLAVE

- Same Version of MySQL?
- Same Tables?
- Same Data?

REPLICATION FAILOVER



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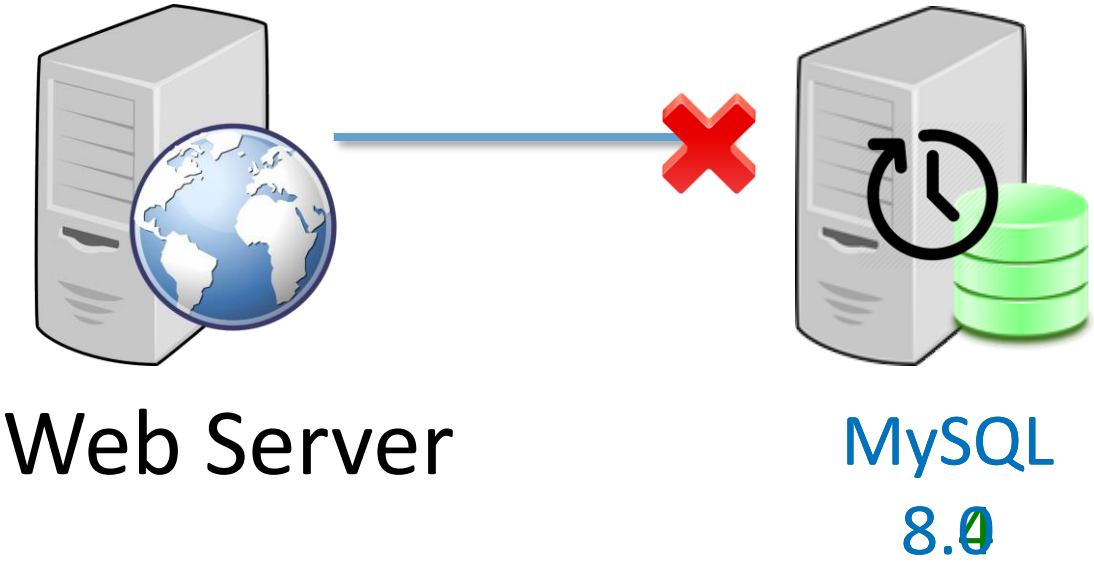
VERSION UPGRADES



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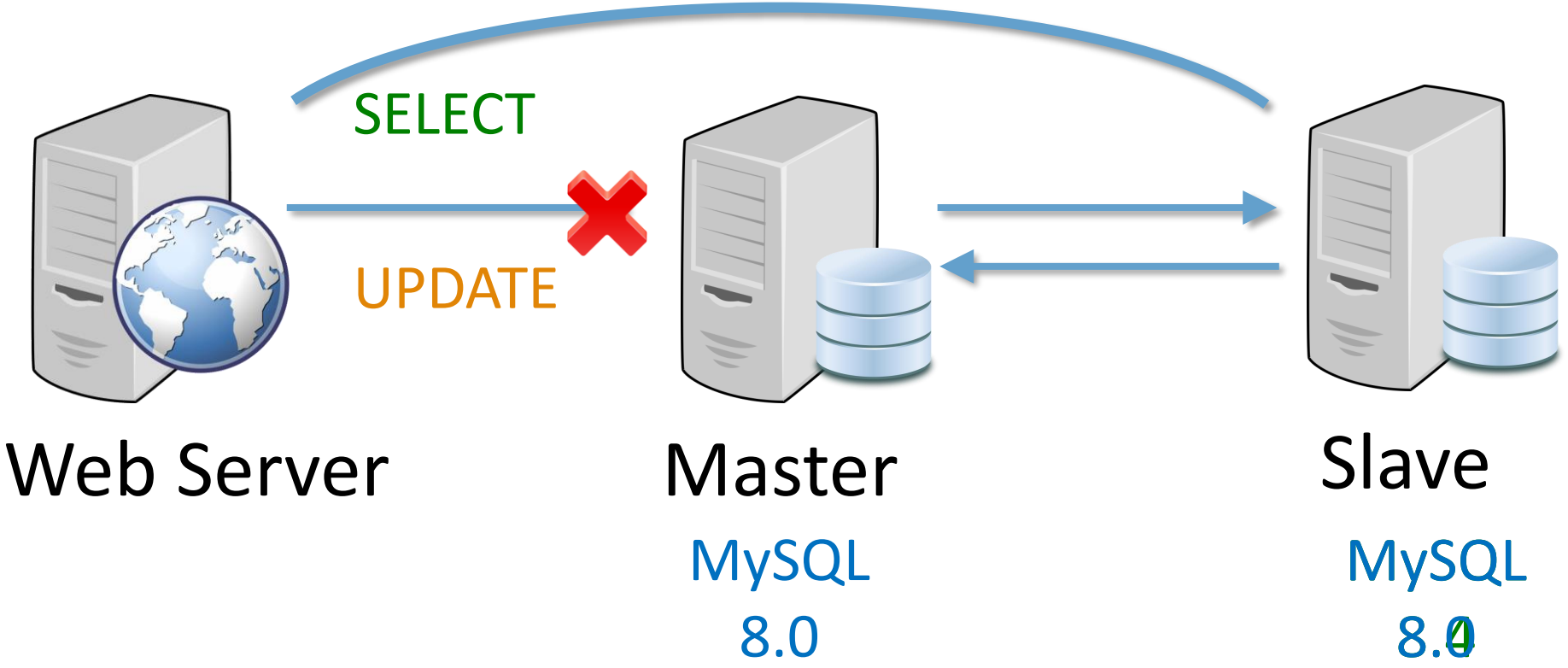
VERSION UPDATES WITH REPLICATION



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